A cutting-plane method for exact-size code bounds, with fifteen new upper bounds for binary codes

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Abstract

Let $A(n,d)$ be the largest size of a binary code of length n and minimum distance d. We give a general method that lowers upper bounds on $A(n,d)$ found by semidefinite programming, and we apply it to many entries with $n \le 28$. It improves fifteen of them, for example $A(17,6) \le 339$, $A(28,8) \le 32148$ and $A(26,6) \le 84243$. The method fixes the size M of the code and adds, by a cutting-plane loop, linear inequalities that are valid for codes of size exactly M: for a subspace V of dimension r, the code meets the 2^r cosets of $V^\perp$ in integer numbers of codewords, and these numbers cannot all be equal when $2^r \nmid M$. It then excludes $M = \lfloor s \rfloor, \lfloor s \rfloor - 1, \ldots$ one at a time, where s is a known bound, usually the SDP value. Every bound has an exact rational certificate.

1 Results

Table 1 lists the new bounds; the old values are those of Brouwer's table [3] in October 2026. Since $A(n,d) = A(n-1,d-1)$ for even d, each line also bounds $A(n-1,d-1)$.

(n,d)	old	new	SDP	added inequalities	premise
$(17,4)$	3276	**3272**	Schrijver	coset-count	–
$(21,4)$	43688	**43681**	Schrijver	coset-count	–
$(25,4)$	599184	**599174**	Schrijver	coset-count	–
$(22,6)$	6941	**6939**	Schrijver	coset-count	–
$(24,6)$	24106	**24102**	Schrijver	coset-count	–
$(26,6)$	84260	**84243**	Schrijver	coset-count	–
$(27,6)$	157285	**157281**	Schrijver	coset-count	–
$(28,6)$	291269	**291265**	Schrijver	coset-count	–
$(28,8)$	32151	**32148**	Schrijver	coset-count	–
$(17,6)$	340	**339**	Schrijver	coset-count, Lemma 5	–
$(18,6)$	673	**672**	GMS	coset-count ($r=1$)	–
$(21,10)$	47	**46**	GMS	coset-count ($r=1$)	–
$(18,4)$	6551	**6543**	Schrijver	coset-count, Lemma 4(a)	$A(17,4) \le 3272$
$(19,4)$	13104	**13085**	Schrijver	coset-count, Lemma 4(a)	$A(18,4) \le 6544$
$(27,4)$	2396736	**2396695**	Schrijver	coset-count, Lemma 4(b)	$A(25,4) \le 599174$

Table 1: New upper bounds. The last three use earlier bounds of this table as premises. Also $A(26,4) \le 2A(25,4) \le 1198348$ (old 1198368) follows at once.

2 Notation and set-up

Let $C \subseteq \mathbb{F}_2^n$ with $|C| = M$ and minimum distance at least d, where d is even. Write $A_i = \frac{1}{M}|\{(c,c') \in C^2 : d(c,c') = i\}|$ for the distance distribution ($A_0 = 1$, $\sum_i A_i = M$), $K_k(i) = \sum_j (-1)^j \binom{i}{j}\binom{n-i}{k-j}$ for the Krawtchouk polynomials, and $S_u = \sum_{c \in C} (-1)^{u \cdot c}$ for $u \in \mathbb{F}_2^n$.

Lemma 1. *For $0 \le k \le n$:* $\sum_{\mathrm{wt}(u)=k} S_u^2 = M \sum_{i=0}^n A_i K_k(i)$.

Proof. $S_u^2 = \sum_{c,c' \in C} (-1)^{u \cdot (c+c')}$, and $\sum_{\mathrm{wt}(u)=k} (-1)^{u \cdot x} = K_k(\mathrm{wt}(x))$. □

Two reductions. (i) If C is a code with the above parameters, delete one coordinate and add a parity bit: the result has the same n, M and d, and all its weights are even. So we may assume that all distances in C are even. (ii) A code with more than M words contains a code with exactly M words. Hence, *to prove $A(n,d) \le U$ it suffices to show that no code of size exactly $M = U + 1$ exists.*

The relaxations. Schrijver's SDP [8] and the four-point SDP of Gijswijt, Mittelmann and Schrijver (GMS) [4] assign to each code C a vector x_C of real variables with these properties: x_C satisfies all constraints of the SDP; the objective value at x_C is $|C|$; and, for fixed $M = |C|$, each A_i is a linear function of x_C. (In Schrijver's SDP, $A_i = \binom{n}{i} x_{i,0}^0$; in the GMS SDP, $A_i = 2^n \binom{n}{i} x(\{v,w\})/M$ for $d(v,w) = i$.) Hence any linear inequality in (A_i) that holds for all codes of size M can be added, and if the maximum of the enlarged SDP is less than M, no code of size M exists. We use constraints of these SDPs that are valid for codes (for GMS a subset of the published blocks), together with the even-distance constraints of reduction (i).

Certificates. Write the SDP as: maximize $c^\top x$ subject to $F_k(x) \succeq 0$, $g_j(x) \ge 0$ and the box $0 \le x \le b$ (with $b = 1$ for Schrijver's SDP and $b = M/2^n$ for GMS), where the F_k are affine maps to symmetric matrices and the g_j are affine. For any $Y_k \succeq 0$ and $z_j \ge 0$, every feasible x satisfies $c^\top x \le L(x) := c^\top x + \sum_k \langle Y_k, F_k(x)\rangle + \sum_j z_j g_j(x)$, and L is affine, so $\max_{0 \le x \le b} L(x)$ is an upper bound. Each of our certificates gives $Y_k = V_k V_k^\top / 4^K$ with an integer K and integer matrices V_k (so $Y_k \succeq 0$ exactly) and rational $z_j \ge 0$; the bound is then computed in exact rational arithmetic. No floating-point step enters the proof.

3 The coset-count inequality

Theorem 2. *Let $V \le \mathbb{F}_2^n$ be a subspace of dimension r and let $\rho \in \{0, \dots, 2^r - 1\}$ with $\rho \equiv M \pmod{2^r}$. Then*

$$\sum_{v \in V \setminus \{0\}} S_v^2 \;\ge\; \rho\,(2^r - \rho).$$

Proof. Choose a basis $u_1, \dots, u_r$ of V. For $g \in \mathbb{F}_2^r$ let n_g be the number of $c \in C$ with $(u_1 \cdot c, \dots, u_r \cdot c) = g$; these are the intersection sizes of C with the cosets of $V^\perp$. For $v = \sum_i a_i u_i$ we have $S_v = \sum_g n_g (-1)^{a \cdot g}$, the Fourier transform of (n_g) on $\mathbb{F}_2^r$. Since $a \mapsto v$ is a bijection $\mathbb{F}_2^r \to V$, Parseval gives $\sum_{v \in V} S_v^2 = 2^r \sum_g n_g^2$. The n_g are 2^r nonnegative integers with sum $M = 2^r q + \rho$; the sum of squares is smallest when ρ of them equal $q+1$ and the others equal q. So $2^r \sum_g n_g^2 \ge 2^r\big(2^r q^2 + 2\rho q + \rho\big) = M^2 + \rho(2^r - \rho)$. Subtract $S_0^2 = M^2$. □

The inequality depends on the arrangement of C in space. We average it over all coordinate permutations to get an inequality in the distance distribution. Let m_k be the number of words of weight k in $V \setminus \{0\}$.

Theorem 3. *Under the hypotheses of Theorem 2,* $\displaystyle\sum_{k=1}^{n} \frac{m_k}{\binom{n}{k}} \sum_{i=0}^{n} A_i K_k(i) \;\geq\; \frac{\rho(2^r-\rho)}{M}$.

Proof. Theorem 2 holds for $\sigma(V)$ for every permutation σ of the coordinates. Average over $\sigma \in S_n$: a fixed v of weight k is mapped onto each word of weight k equally often, so the average of $S^2_{\sigma(v)}$ is $\binom{n}{k}^{-1} \sum_{\mathrm{wt}(u)=k} S_u^2$. Apply Lemma 1 and divide by M. □

The row is linear in (A_i) and depends only on M and on the weight distribution of V. For $r = 1$ and M odd it says $\sum_i A_i K_k(i) \geq \binom{n}{k}/M$: every S_u is odd. The cases $r \leq 2$ are essentially classical parity arguments (Best et al. [2]; see also [1, 5]); the rows with $r \geq 3$ are, as far as we know, new, and they are the ones that matter for most of our bounds. Section 5 explains how we choose the subspaces.

4 Inequalities that use a premise

Lemma 4. *Let* $|C| = M$.

(a) If $A(n-1,d) \leq P$, *then* $M \leq 2P$ *and* $M \sum_i A_i K_1(i) \leq n(2P-M)^2$.

(b) For $1 \leq k < n$: $\sum_i \binom{n-i}{k} A_i \leq \binom{n}{k} A(n-k,d)$.

Proof. (a) Fix a coordinate j and let a and $M-a$ be the numbers of codewords with $c_j = 0$ and $c_j = 1$. Each class, with coordinate j deleted, is a code of length $n-1$ and distance $\geq d$, so $a \leq P$ and $M - a \leq P$. Hence $M \leq 2P$ and $|S_{e_j}| = |2a - M| \leq 2P - M$. Sum $S^2_{e_j}$ over j and apply Lemma 1 with $k = 1$. (b) Count triples (c, c', K) with $c, c' \in C$, K a k-set of coordinates, and c, c' equal on K. For fixed (c, c') there are $\binom{n-d(c,c')}{k}$ such K. For fixed (c, K) the words c' that equal c on K, with K deleted, form a code of length $n-k$ and distance $\geq d$, so there are at most $A(n-k,d)$ of them. Hence $M \sum_i \binom{n-i}{k} A_i \leq M\binom{n}{k} A(n-k,d)$. □

Both parts are standard counting arguments; cf. [2]. The next row needs no premise.

Lemma 5 (cf. [6])**.** *If* $d = 2h$ *and* $q = \lfloor n/h \rfloor$, *then* $A_{n-h} + q \sum_{i=0}^{h-1} A_{n-i} \leq q$.

Proof. Fix $c \in C$ and let $\bar{c}$ be its complement; $d(c, y) = n - i$ means $d(\bar{c}, y) = i$. At most one codeword lies at distance $< h$ from $\bar{c}$ (two such would be closer than $2h$). If one, y, does, then a codeword x with $d(\bar{c}, x) = h$ would give $d(x, y) < 2h$; so there is none. Otherwise, the codewords at distance exactly h from $\bar{c}$ differ from $\bar{c}$ on h-sets that are pairwise disjoint (their mutual distances are $\geq 2h$), so there are at most q of them. In both cases (number at distance $n-h$ from c) $+\, q\cdot$(number at distance $> n-h$ from c) $\leq q$. Average over c. □

5 The method

Input: (n, d) with d even; a relaxation R (Schrijver's or the GMS SDP with the even-distance constraints); optionally premises, i.e. known bounds on $A(n-k,d)$, used through Lemma 4.

1. Let s be the value of R (or a known bound such as $2A(n-1,d)$), and set $M := \lfloor s \rfloor$.
2. *Exact size.* Add to R the rows of Theorem 3 for all *block subspaces* $V = \langle 1_{T_1}, \ldots, 1_{T_r} \rangle$ with nonempty pairwise disjoint $T_b \subseteq \{1, \ldots, n\}$ and $\rho \neq 0$; for these, $\sum_k m_k z^k = \prod_b (1 + z^{|T_b|}) - 1$. Optionally add the rows of Lemmas 4 and 5.

3. *Separation.* Solve. At the optimum let $\beta_k = M\sum_i A_i K_k(i)/\binom{n}{k}$ (for a real code this is the average of S_u^2 over $\mathrm{wt}(u) = k$, by Lemma 1). The row of V is violated exactly when $\sum_k m_k(V)\beta_k < \rho(2^r - \rho)$. Search for such V by local search over $r \times \ell$ generator matrices of rank r ($2 \le r \le 10$, up to 12 at some entries), stored as the multiset of their nonzero columns, add the violated rows, and solve again. Stop when the value is below M or no violated V is found.

4. *Certificate.* If the value is below M, round the dual solution to rationals, write the PSD multipliers as $V_k V_k^\top/4^K$, and compute the bound exactly (Section 2). If it is below M, the size M is excluded; set $M := M - 1$ and go to step 2. Otherwise stop.

Output: if M_0 is the last excluded size, then $A(n,d) \le M_0 - 1$ by reduction (ii). Runs with Schrijver's SDP started in double precision (CSDP); the GMS runs and all runs with $n \ge 26$ used double-double precision (SDPA-DD [7]), and $(21, 10)$ needed quad-double precision (SDPA-QD). The numerics never enter the proof. (For $(24, 6)$ the sizes 24107 and 24106 were first excluded by classical parity rows with a case split; the proof needs only the certificate for $M = 24103$.)

We ran this procedure on many entries with $n \le 28$, in particular on those whose upper bound in the table comes from linear or semidefinite programming. It improved the fifteen entries of Table 1. The descent stops where the relaxation becomes feasible at M; for instance at $(26, 6)$ and $(28, 8)$ it is feasible at $M = U$ and the separation finds no violated row.

6 Proofs of the bounds

By reduction (ii), each bound $A(n,d) \le U$ in Table 1 follows from one certificate showing that the SDP with the added rows, for $M = U + 1$, has maximum less than M. Table 2 gives the certified bounds.

(n,d)	$M = U+1$	certified bound	SDP value without added rows ($\approx$)
$(17,4)$	3273	3272.153	3276.8
$(21,4)$	43682	43681.452	43690.667
$(25,4)$	599175	599173.716	599186.288
$(22,6)$	6940	6939.701	6943.695
$(24,6)$	24103	24102.898	24107.886
$(26,6)$	84244	84243.831	84260.571
$(27,6)$	157282	157281.944	157286.400
$(28,6)$	291266	291265.483	291271.111
$(28,8)$	32149	32148.878	32151.796
$(17,6)$	340	339.581	341.333 (with Lemma 5)
$(18,6)$	673	672.883	673.005 (GMS)
$(21,10)$	47	46.916	47.007 (GMS)
$(18,4)$	6544	6543.538	–
$(19,4)$	13086	13085.663	–
$(27,4)$	2396696	2396695.99997	–

Table 2: The decisive certificates. A certified bound is an exact rational upper bound on the maximum of the SDP with the added rows, here rounded up; the last column is for orientation only and is not used in the proofs.

Bounds without premise. For the first nine lines the SDP is Schrijver's with coset-count rows only. For $(17, 6)$ we also add Lemma 5, which reads $A_{14} + 5A_{16} \leq 5$ (odd A_i vanish). Both kinds of rows are needed: without the coset-count rows the maximum is 341.33, and without Lemma 5 the solver gives about 340.00005 at $M = 340$. For $(18, 6)$ and $(21, 10)$ the size M is odd and the GMS SDP with the $r = 1$ rows suffices.

Bounds that use a premise. Splitting a code by one coordinate gives $A(n, d) \leq 2A(n - 1, d)$, so $A(18, 4) \leq 2 \cdot 3272 = 6544$; Lemma 4(a) with $P = 3272$ excludes $M = 6544$. In the same way Lemma 4(a) with $P = 6544$ (valid since $A(18, 4) \leq 6544$) excludes $M = 13086$. Doubling twice gives $A(26, 4) \leq 1198348$ and $A(27, 4) \leq 2396696$; Lemma 4(b) for $k = 1, 2$ with premises $A(26, 4) \leq 1198348$ and $A(25, 4) \leq 599174$ excludes $M = 2396696$.

7 Remarks

Numerics and checking. For $n \geq 26$ CSDP reported spurious feasibility, and double-double precision was needed. Most certificates were also checked by a second program written independently from Schrijver's formulas; for $(21, 10)$ the independent check was done on a second certificate from a different implementation of the GMS SDP.

Sanity check. The same programs do not exclude the sizes of known codes (e.g. $(27, 81920, 6)$ and $(28, 139264, 6)$). A larger gain than ours needs a stronger relaxation.

Data. The decisive certificates, the exact checkers and the generator programs are in the data record of this paper (`https://doi.org/10.5281/zenodo.23599648`).

Use of AI. AI tools were used in the computations and in the writing of this paper. The authors do not guarantee the correctness of the results.

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